Literature dissection

Spring 2026

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2026-05-29

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Photo documentation

Table

Photo point number	Date of photo	Season	Direction of photo	Link to photo
19	October 2023	Fall	West	Link is here
19	October 2023	Fall	North	Link is here
19	January 2024	Winter	West	Link is here
19	January 2024	Winter	North	Link is here
19	April 2024	Spring	West	Link is here
19	April 2024	Spring	North	Link is here

We chose to examine water year 2024. This water year is relevant to our study because it is the year with the most precipitation in our study. We chose photo point 19. This point shows the water from Venoco Bridge. It is relevant to our study as it shows the water levels and this is a factor in waterfowl abundance. Changes we see in the photos include water level changes, the occasional bird, and the beginning of the construction of the Cove.

Reports

Report

Title: Bird Community Analysis Report 1967–2015

Authors: Stahlheber, Karen A

Year published: 2015

Permalink: https://escholarship.org/uc/item/1sg3v19d

Dataset used (likely): `birds.csv`

Key insights: This report analyzed nearly 50 years of bird survey data (1967–2014) across wetland and upland habitats in the Goleta area to characterize bird community composition, diversity, abundance, and habitat use patterns. The study found that bird communities differed among habitat types and locations, highlighting the importance of wetland habitat characteristics in shaping local bird abundance and diversity.

Relevance to your study (be specific here with the question you have asked, the figures you've made, the insights you've gained, etc.): This report is relevant because it provides long-term baseline data on bird abundance and diversity in local wetland habitats, including areas near North Campus Open Space. Understanding historical patterns in waterbird communities helps place any observed relationships between precipitation and waterfowl abundance into a broader ecological context.

Annotated bibliography

Paper 1

Title: Trends in abundance of wintering waterbirds relative to rainfall patterns at a central California estuary, 1972-2015

Authors: Lynne E. Stenzel and Gary W. Page

Main findings: They found that it was more common than not to find a trend between abundance and rainfall. 14 of the 19 taxa sampled were found to have a negative relationship between species abundance and the current year's precipitation. A negative relationship between species abundance and the previous year's precipitation was more common than positive. (Stenzel and Page 2011)

Relevance to study: This paper is relevant to our study because we are focusing on how precipitation affects waterfowl abundance at NCOS. The paper focuses on waterbirds which is similar to waterfowl, just more broad. This paper also found results similar to what we found.

Paper 2

Title: Habitat quality and drought effects on breeding mallard and other waterfowl populations in California, USA

Authors: Sharon N. Kahara, Daniel Skalos, Buddhika Madurapperuma, Kaitlyn Hernandez

Main findings: This study found that drought and habitat quality had an effect on waterfowl population. Drought proved to be one of the most influential factor on waterfowl abundance. It found that generally, drought decreased population in some regions but not in all, finding that in some areas it increased. (Kahara et al. 2021)

Relevance to study: This is relevant to our study because drought is closely tied to precipitation. While this is in central California, the influence of drought on population may be seen in our study.

Paper 3

Title: Forecasting waterfowl population dynamics under climate change – Does the spatial variation of density dependence and environmental effects matter?

Authors: Qing Zhao, Emily Silverman, Kathy Fleming, G. Scott Boomer

Main findings: This report found that due to climate change, Mallard density is predicted to decline. One of the climate variables they examine is precipitation. (Zhao et al. 2016)

Relevance to study: This is relevant to our study because we are interested in the effects of precipitation on waterfowl, such as the Mallard. Their modelling is similar to what we are retroactively looking at.

Paper 4

Title: The effect of climate change on optimal wetlands and waterfowl management in Western Canada

Authors: Patrick Withey, G. Cornelis van Kooten,

Main findings: This article finds that precipitation is strongly correlated with wetland habitat availability. The modelling they used predicted that climate change could decrease wetlands by 7-47%. They also note that precipitation from the year prior is a good indicator of habitat avalibility. (Withey and Kooten 2011)

Relevance to study: This paper is relevant to our study because they are focusing on how climate change variables such as precipitation and temperature changes will affect wetland habitats. this is where waterbirds live and precipitation affects the quality of the habitat.

Paper 5

Title: Projected Impacts of Climate, Urbanization, Water Management, and Wetland Restoration on Waterbird Habitat in California's Central Valley

Authors: Elliot L. Matchett, Joseph P. Fleskes

Main findings: Climate change, urbanization, and changes in water management are likely to reduce waterbird habitat in California's Central Valley over the coming decades. Planned wetland restoration can offset many of these losses through about 2065, but it becomes insufficient under most future scenarios after that point. The greatest habitat declines occur under warmer, drier climate conditions combined with reduced water availability, suggesting that additional wetland restoration and climate adaptation strategies will be needed to support waterbird populations in the future. (Matchett and Fleskes 2017)

Relevance to study: It directly links precipitation/climate change and wetland restoration to waterbird habitat in California. Finds that warmer, drier conditions significantly reduce waterbird habitat area.

Paper 6

Title: Long-Term Changes in the Abundance of Benthic Foraging Birds in a Restored Wetland

Authors: Lucas Mander, Luca Scapin, Rodney M. Foster, Chris B. Thaxter, Niall H. K. Burton

Main findings: The study found that managed realignment successfully created new intertidal habitat that initially supported increasing numbers of waterbirds. However, as sediment accumulated and site elevations increased over time, the abundance of several waterbird species declined. These results suggest that while habitat restoration can benefit waterbirds, long-term environmental changes can affect habitat suitability and should be considered in restoration planning. (Mander et al. 2021)

Relevance to study: This study is relevant because it demonstrates how changes in wetland habitat conditions can influence waterbird abundance over time. Since precipitation affects water levels, flooding extent, and habitat availability at North Campus Open Space, it may similarly shape the abundance and distribution of waterfowl in the area.

Paper 7

Title: The Impact of Drought on Waterbirds and Their Wetland Habitats in California's Central Valley

Authors: Joseph Fleskes, Matthew Reiter

Main findings: The study found that wintering waterfowl responded strongly to restored Wetlands Reserve Program sites, with bird use increasing substantially after restoration and remaining high for years. Restored wetlands were especially effective when located near flooded rice fields and in areas with high existing waterfowl abundance. These results suggest that wetland restoration is an effective strategy for increasing waterfowl habitat in California's Central Valley.(Fleskes et al. 2019)

Relevance to study: It evaluates how drought affects waterbird distribution, abundance, and wetland habitat timing in California. This closely mirrors our study's question.

Paper 8

Title: Waterfowl–habitat associations during winter in an urban North Atlantic estuary

Authors: Richard A. McKinney, Scott R. McWilliams, Michael A. Charpentier

Main findings: The study found that wintering waterfowl were generally more abundant in areas with more surrounding vegetated upland habitat and less nearby residential development. Hunting activity also reduced species richness, while habitat and landscape characteristics together influenced where waterfowl were most likely to occur. These findings suggest that both local habitat conditions and broader landscape development patterns are important for waterfowl conservation in urban estuaries. (McKinney et al. 2006)

Relevance to study: It examines factors affecting waterfowl abundance and species richness at an estuarine site. This provides a similar coastal wetland context to NCOS and finds that habitat characteristics and landscape composition are key drivers.

Paper 9

Title: Waterbird Use of a Urban Stormwater Wetland System in Central California, USA

Authors: Joan M. Duffield

Main findings: The study found that waterbird abundance and species diversity varied significantly between seasons and between a stormwater wetland and natural marsh. Diving Ducks preferred the the natural marsh, while shorebirds preferred the stormwater wetland. Duffield (1986)

Relevance to study: This is directly relevant to our study because water availability is directly tied to precipitation and linked to bird abundance. This is a very similiar relationship to the one we are analyzing. Duffield found that the areas that received the most direct stormwater, attracted the greatest waterbird use, which draws parallels from our own hypothesis.

Paper 10

Title: Consequences of rainfall variation for breeding wetland blackbirds

Authors: Robert J Fletcher, Jr. Rolf R Koford

Main findings: During dry years, yellow headed blackbirds, showed a marked reduction in density and a complete reproductive failure. The failure was attributed primarily to nest predation which was negatively correlated with water levels in wetlands. (Fletcher, Jr. and Koford 2004)

Relevance to study: This paper is directly relevant to our own study because it is similiarly analyzing how rainfall variation can affect wetland bird demography. Rain is also being used to study population dynamics which is mirrors our own research.

Paper 12

Title: The Effects of Seasonal Flooding on Seed Availability for Spring Migrating Waterfowl

Authors: Andrew K. Greer, Bruce D. Dugger, Dave A. Graber, Mark J. Petrie

Main findings: The main findings of this study were that fall-flooded wetlands had much greater overwinter seed loss, compared to spring flooded wetlands, meaning that there was less food available by the time spring migrants arrived. The authors recommend spring flooding for wetland managers wanting to maximize food availability for migrating fowls.(GREER et al. 2007)

Relevance to study: This paper is relevant because it provides a relevant natural history background looking at how wetland water management affects food availability for waterfowls. Although NCOS is a natural system, I think the insight from the paper is still relevant looking at how wetter conditions affects foraging opportunities and ultimately that is tied to abundance.

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